



Student Number : \_\_\_\_\_

# YEAR 12 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2021

## Mathematics Extension 1

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### General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper (pages 21-24)
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

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### Total marks: 70

#### Section I – 10 marks (pages 2 – 8)

- Attempt Questions 1 – 10
- Answer on the multiple choice sheet provided on page 19 (you may detach this sheet)
- Allow about 15 minutes for this section

#### Section II – 60 marks (pages 9 – 18)

- Attempt Questions 11 – 14
- Answer each question in a separate answer booklet
- Allow about 1 hours and 45 minutes for this section

**Section 1 – 10 Marks (1 mark each)**

- Attempt Questions 1-10
  - Answer on the Multiple Choice Sheet provided on Page 19  
(You may detach this sheet)
  - Allow about 15 minutes for this section
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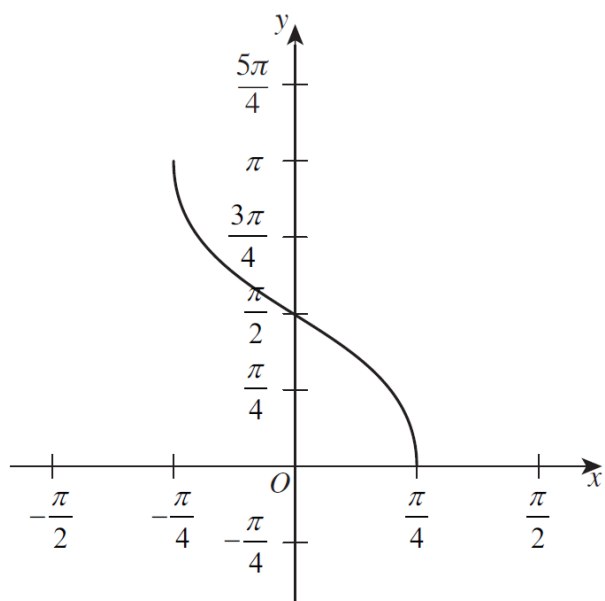
1. Which of the following is NOT a factor of the polynomial  
 $P(x) = x^3 - x^2 - 8x + 12$  ?

- (A)  $(x + 3)$
- (B)  $(x + 3)^2$
- (C)  $(x - 2)$
- (D)  $(x - 2)^2$
- 

2. What is the primitive of  $\frac{\ln 2}{\sqrt{\pi - x^2}}$  ?

- (A)  $(\ln 2)(x) \left( \sin^{-1} \left( \frac{x}{\sqrt{\pi}} \right) \right) + C$
- (B)  $(\ln 2) \left( \sin^{-1} \left( \frac{x}{\sqrt{\pi}} \right) \right) + C$
- (C)  $\left( \frac{\ln 2}{\sqrt{\pi}} \right) \left( \sin^{-1} \left( \frac{x}{\sqrt{\pi}} \right) \right) + C$
- (D)  $(\ln 2) \left( \sin^{-1} \left( \frac{x}{\pi} \right) \right) + C$

3. The graph of an inverse trigonometric function is shown.



What is the equation of this function?

(A)  $\frac{4}{\pi} \cos^{-1} x$

(B)  $\frac{\pi}{4} \cos^{-1} x$

(C)  $\cos^{-1} \left( \frac{4x}{\pi} \right)$

(D)  $\cos^{-1} \left( \frac{\pi x}{4} \right)$

4. A bag contains 2 red, 5 blue, 6 white, 11 green and 14 yellow marbles. What is the minimum number of marbles that need to be chosen randomly from the bag to ensure that 6 marbles of the same colour have been chosen?

(A) 16

(B) 17

(C) 23

(D) 34

- 
5. The function  $f(x) = x^2 - 4x + 7$  has an inverse function  $f^{-1}(x)$  in the domain  $x \geq 2$ .

What is the value of  $f^{-1}(f(m))$ ,  
where  $m$  is a real number NOT in the domain  $x \geq 2$  ?

(A)  $m$

(B)  $2 - m$

(C)  $2 + m$

(D)  $4 - m$

6. The vectors  $4\hat{i} - \hat{j}$  and  $3\hat{i} + m\hat{j}$  are perpendicular.  
What is the value of  $m$ ?

- (A)  $-12$   
(B)  $-1$   
(C)  $3$   
(D)  $12$

- 
7. A piece of hot metal is placed in a room with a surrounding air temperature of  $15^{\circ}\text{C}$  and is allowed to cool.  
It loses heat according to Newton's Law of Cooling:

$$\frac{dT}{dt} = -k(T - A) \text{ where}$$

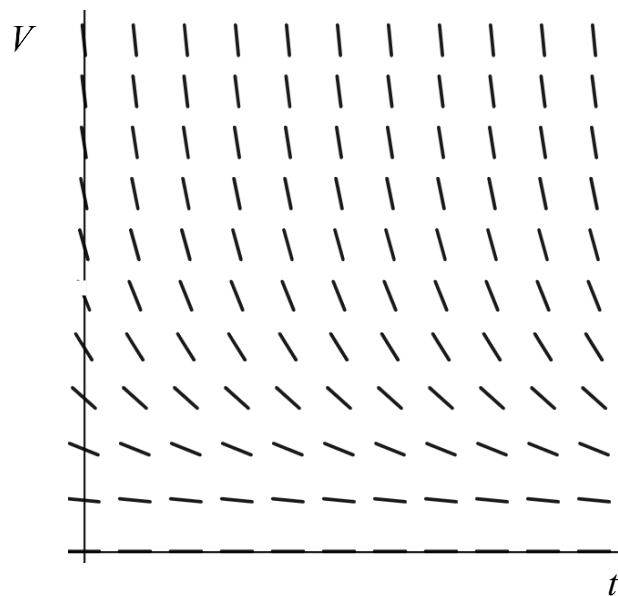
- $T$  is the temperature of the metal in  $^{\circ}\text{C}$ , at time  $t$  minutes
- $A$  is the surrounding air temperature
- $k$  is a positive constant

After 5 minutes the temperature of the metal is  $75^{\circ}\text{C}$   
and after a FURTHER 3 minutes the temperature of the metal is  $45^{\circ}\text{C}$ .

What is the value of  $k$  in the above equation?

- (A)  $3\ln(0.5)$   
(B)  $3\ln(2)$   
(C)  $\frac{\ln(0.5)}{3}$   
(D)  $\frac{\ln(2)}{3}$

8. A direction field for the volume of water,  $V$  megalitres, in a reservoir  $t$  years after 2021 is shown below.



According to this model, for  $k > 0$ , what is the value of  $\frac{dV}{dt}$  ?

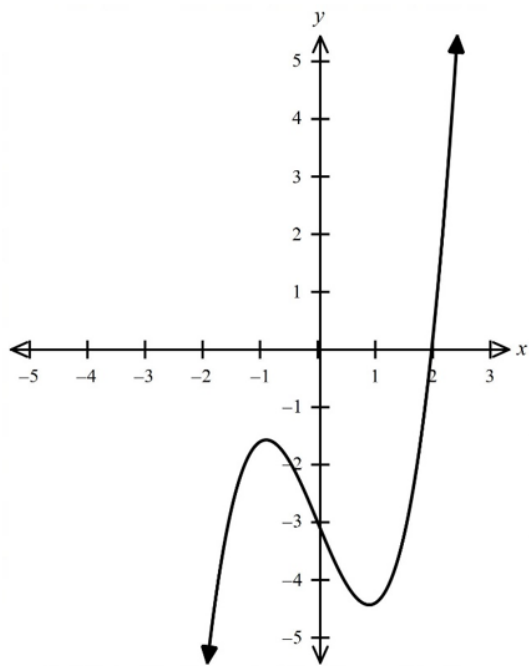
(A)  $-kV^2$

(B)  $-\frac{k}{V}$

(C)  $\frac{k}{V}$

(D)  $kV^2$

9. Consider the graph of  $y = f(x)$  shown below:



Which of the following would have two MORE roots than  $f(x)$ ?

(A)  $y = -2 \times f(x)$

(B)  $y = f(x) + 3$

(C)  $y = f^{-1}(x)$

(D)  $y = f(x + 3)$

10. It is known that  $\sin x = \frac{1}{3}$ , where  $\frac{\pi}{2} < x < \pi$

What is the value of  $\cos(2x)$  ?

(A)  $-\frac{7}{9}$

(B)  $\frac{7}{9}$

(C)  $\frac{2}{3}$

(D)  $-\frac{4\sqrt{2}}{3}$

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**END OF SECTION 1**

## SECTION 2 – 60 Marks

- Attempt Questions 11-14
- Allow about 1h 45min for this section
- Answer each question in a separate writing booklet (extra writing booklets are available upon request from the examination supervisor)
- Show all relevant mathematical reasoning and/or calculations

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Marks

**Question 11** (15 marks) Use a SEPARATE writing booklet

(a) (i) Write  $\cos(7x)\sin(5x)$  as a sum/difference of trigonometric ratios 1

(ii) Hence, solve  $2\cos(7x)\sin(5x) = \sin(12x) - \frac{\sqrt{3}}{2}$  for  $0 \leq x \leq 2\pi$  3

(b) Prove by mathematical induction that  $2^{2n} + 6n - 1$  is divisible by 3 for all integers  $n \geq 1$  3

(c) Consider the polynomial  $P(x) = x^3 - 2px + q$  where  $p$  and  $q$  are constants and  $p \neq 0$ .

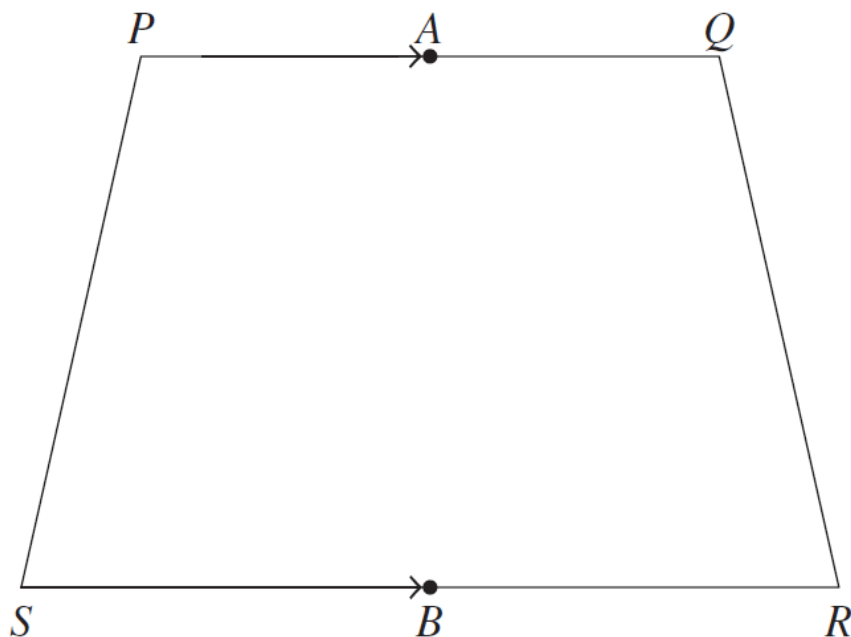
It is given that  $\alpha$ ,  $\beta$  and  $\alpha + \beta$  are roots of the equation  $P(x) = 0$ .

(i) Prove that  $\alpha = -\beta$  1

(ii) Hence, or otherwise, find all the roots of  $P(x) = 0$  in terms of  $p$ . 3

**Question 11 continues on page 10**

- (d)  $PQRS$  is a trapezium with  $A$  and  $B$  being the midpoints of the parallel sides  $PQ$  and  $RS$  respectively. Marks



Let  $\overrightarrow{PA} = \underline{a}$  and  $\overrightarrow{SB} = \underline{b}$ .

- (i) Express  $\overrightarrow{QR}$  in terms of  $\underline{a}$ ,  $\underline{b}$  and  $\overrightarrow{AB}$

2

- (ii) Hence, or otherwise, show that  $\overrightarrow{AB} = \frac{1}{2}(\overrightarrow{PS} + \overrightarrow{QR})$

2

**END OF QUESTION 11**

**Question 12** (15 marks) Use a SEPARATE writing booklet

Marks

- (a) (i) Write the expression  $\sqrt{3}\sin x + \cos x$  in the form  $R\sin(x + \alpha)$ ,  
 where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$

2

- (ii) Write down the three missing numbers that would appear in the cells of the table below, where  $g(x) = \sqrt{3}\sin x + \cos x = R\sin(x + \alpha)$  and the numbers are given correct to 2 decimal places

1

| $x$     | $-\frac{\pi}{6}$ | 0 | $\frac{2\pi}{6}$ | $\frac{5\pi}{6}$ | $\frac{8\pi}{6}$ | $\frac{11\pi}{6}$ | $\frac{14\pi}{6}$ | $\frac{17\pi}{6}$ | $\frac{20\pi}{6}$ |
|---------|------------------|---|------------------|------------------|------------------|-------------------|-------------------|-------------------|-------------------|
| $\ln x$ |                  |   | 0.05             | 0.96             | 1.43             | 1.75              |                   |                   | 2.35              |
| $g(x)$  | 0                | 1 | 2                | 0                | -2               | 0                 | 2                 | 0                 | -2                |

- (iii) Sketch the graphs of  $y = \ln x$  and  $y = g(x)$  on the same number plane, showing clearly any point(s) of intersection.  
 (You do not need to give the coordinates of any point(s) of intersection, just show their location on your sketch)

3

- (iv) Hence, determine the number of solutions to the equation  $\sqrt{3}\sin x + \cos x = \ln x$  in the domain  $(0, \infty)$

1

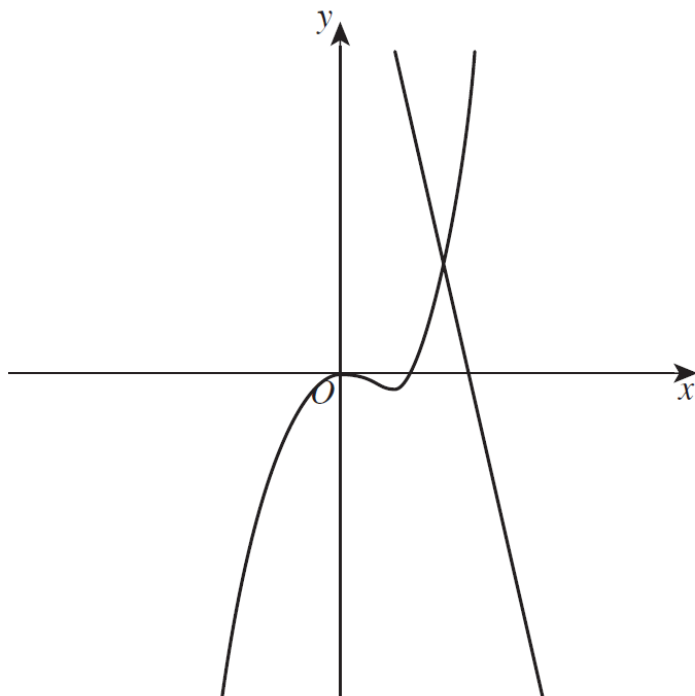
- (b) Use the identity  $(1+x)^m(1+x)^n = (1+x)^{m+n}$  to show that

3

$$\binom{m+n}{4} = \binom{n}{4} + \binom{n}{3}\binom{m}{1} + \binom{n}{2}\binom{m}{2} + \binom{n}{1}\binom{m}{3} + \binom{m}{4}$$

**Question 12 continues on page 12**

- (c) The graphs of  $g(x) = mx + b$  and  $f(x) = 3x^3 - 2x^2$  are shown below



The solution to the equation  $mx + b < 3|x|^3 - 2|x|^2$  is  $x \in (-\infty, -2) \cup (1, \infty)$ .

- (i) Sketch the graph of  $y = 3|x|^3 - 2|x|^2$  1
- (ii) Hence, find the values of  $m$  and  $b$ . 3
- (d) In how many different ways can 6 runners finish 1<sup>st</sup>, 2<sup>nd</sup> and 3<sup>rd</sup> in a race? 1  
(assume no equal placings)

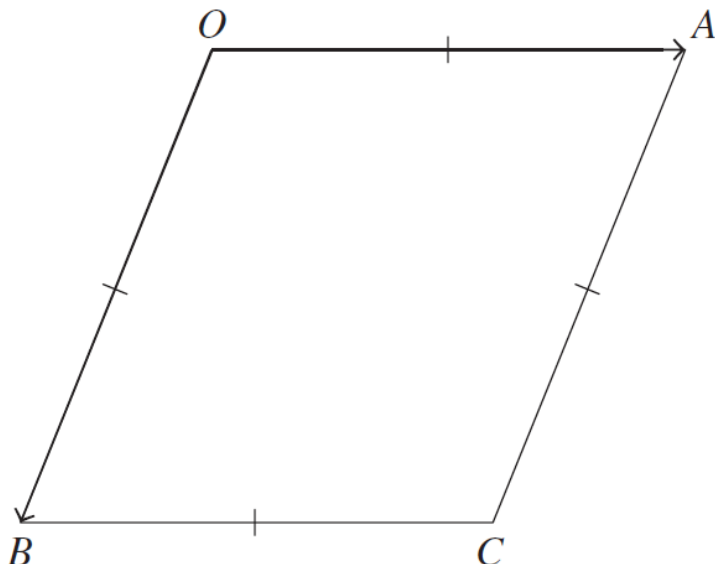
**END OF QUESTION 12**

**Question 13** (15 marks) Use a SEPARATE writing booklet

Marks

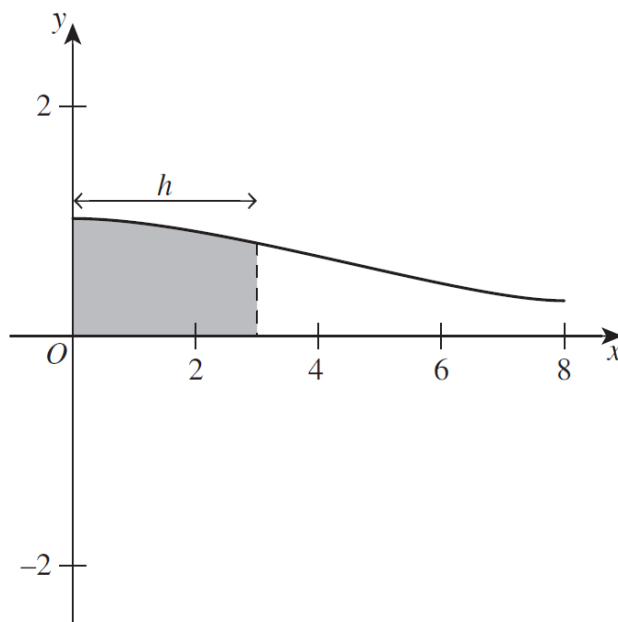
- (a)  $OACB$  is a rhombus with  $\overrightarrow{OA} = \underline{a}$  and  $\overrightarrow{OB} = \underline{b}$

2



Using vector methods, prove that the diagonal  $OC$  bisects angle  $AOB$ .

- (b) The graph of  $y = \frac{3}{\sqrt{9+x^2}}$  for  $x > 0$  is shown below.

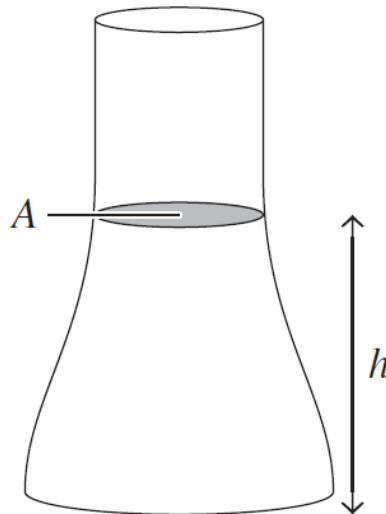


**Question 13(b) continues on page 14**

- (i) The area bounded by  $y = \frac{3}{\sqrt{9+x^2}}$ ,  
the  $x$ -axis and the lines  $x=0$  and  $x=h$  is rotated about the  $x$ -axis  
to create a solid of revolution. 2

Show that the volume of the solid is given by  $V = 3\pi \tan^{-1}\left(\frac{h}{3}\right)$  cubic units.

- (ii) The solid of revolution of the graph has the shape of a vase lying down  
on its side. 2  
The diagram shows a vase of the same shape standing upright.



As water is being poured into the vase, the height  $h$  of the water  
in the vase is increasing at a rate of  $3 \text{ cm/s}$ .

Find the exact rate at which the volume of water in the vase,  
 $V$ , is increasing when  $h = 6$ .

- (iii) The surface of the water forms a circular shape with area  $A$ .  
By using the original graph, or otherwise, find the exact rate at which the  
area,  $A$ , is decreasing when  $h = 6$ . 2

**Question 13 continues on page 15**

- (c) The spread of flu through a student population is modelled by the equation  

$$S = \frac{2000}{1 + 199e^{-0.4t}}$$
 where  $S$  is the total number of students infected after  $t$  days

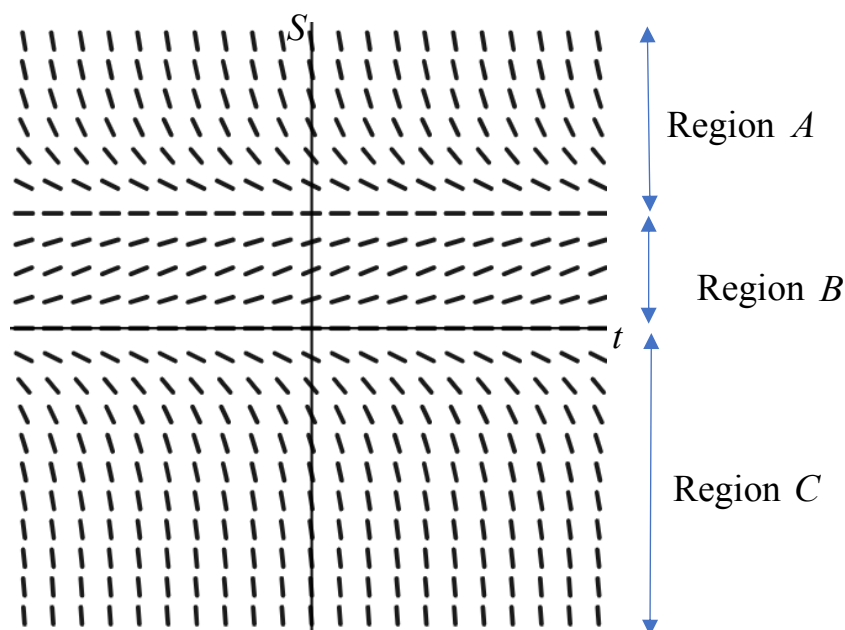
- (i) Show that the given equation for  $S$  satisfies the differential equation

$$\frac{dS}{dt} = \frac{S}{5} \left( 2 - \frac{S}{1000} \right)$$

3

- (ii) The slope field of  $\frac{dS}{dt} = \frac{S}{5} \left( 2 - \frac{S}{1000} \right)$  is shown.

2



There are 3 regions labelled  $A$ ,  $B$  and  $C$  in which a solution curve can be found.

In which one of these three regions can the solution curve exist?  
 Justify your answers with reference to constant solutions and the initial value of  $S$ .

- (iii) According to this model, after how many days does the rate of increase reach a maximum? Give your answer to the nearest number of days.

2

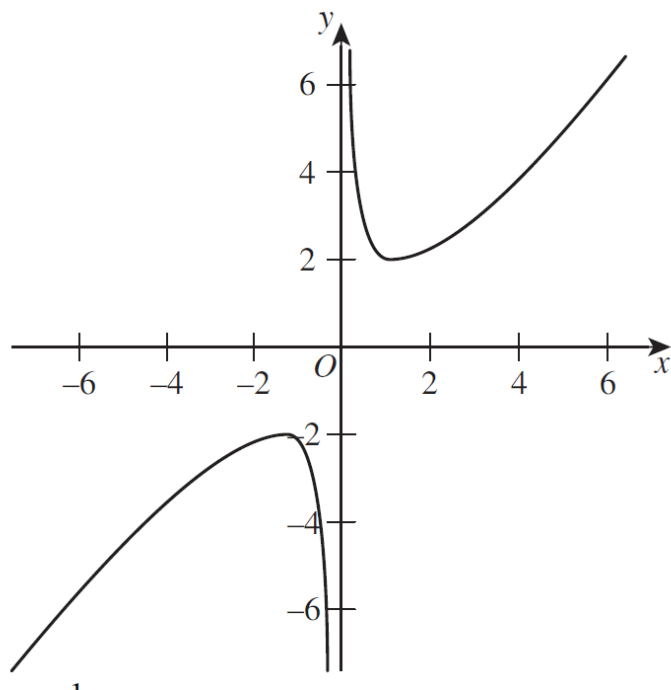
**END OF QUESTION 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet

Marks

- (a) The graph of the function  $f(x) = \frac{x^2 + 1}{x}$  is shown

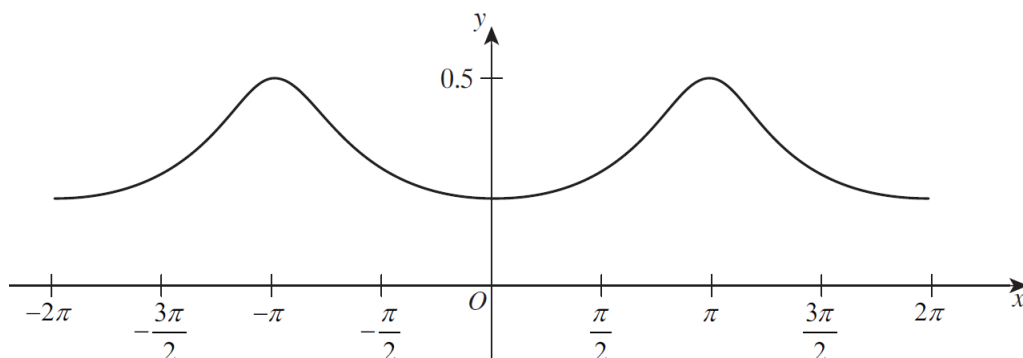
3



Sketch the graph of  $y = \frac{1}{\sqrt{f(x)}}$  showing all important features including turning point(s), intercept(s) and asymptote(s).

**Question 14 continues on page 17**

- (b) The graph of the function  $y = \frac{1}{5 + 3\cos x}$  for  $-2\pi \leq x \leq 2\pi$  is shown below:



- (i) By using the substitution  $t = \tan\left(\frac{x}{2}\right)$  and  $t$  – formula(s), show that 3

$$\int \frac{1}{5 + 3\cos x} dx = \frac{1}{2} \tan^{-1}\left(\frac{1}{2} \tan\left(\frac{x}{2}\right)\right) + C \text{ where } C \text{ is a constant.}$$

- (ii) Hence, show that the area bounded by the curve  $y = \frac{1}{5 + 3\cos x}$ , 2  
the lines  $x = 0$ ,  $x = \pi$  and the  $x$  – axis is  $\frac{\pi}{4}$

You may use the substitution  $\tan \frac{\pi}{2} = \infty$  if necessary.

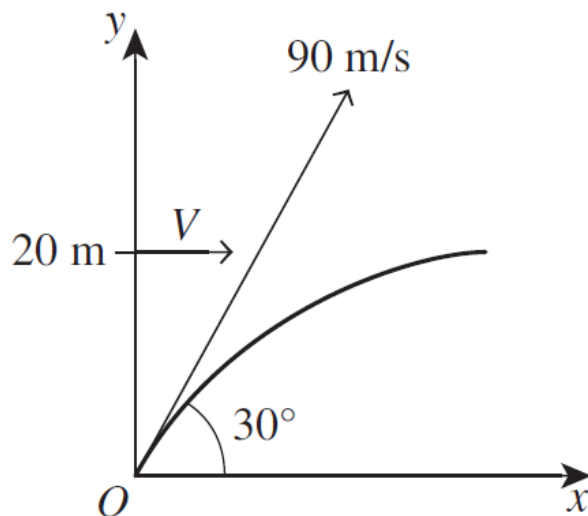
**Question 14 continues on page 18**

- (c) (i) Initially a golf ball was hit from the ground at  $90 \text{ ms}^{-1}$  at an angle of  $30^\circ$  to the horizontal and  $g = 10 \text{ ms}^{-2}$ . 2

Show that the position vector of the golf ball after  $t$  seconds is given by

$$\underline{s}(t) = (45\sqrt{3}t)\underline{i} + (45t - 5t^2)\underline{j}$$

- (ii) Five seconds after the golf ball was hit, a small stone was fired with velocity  $V$  horizontally from a point  $20 \text{ m}$  above the ground. 3



By finding the position vector of the stone, or otherwise, show that the two objects collide after the golf ball has travelled for 21 seconds.

- (iii) What is the stone's speed at collision?  
Give your answer to 3 significant figures. 2

**END OF EXAMINATION**

$$1/ \quad P(2) = (2)^3 - (2)^2 - 8(2) + 12 = 0$$

$\therefore (x-2)$  is a factor

$$P(-3) = (-3)^3 - (-3)^2 - 8(-3) + 12 = 0$$

$\therefore (x+3)$  is a factor

Product of roots is  $-12$

$$2^2 \times (-3) = -12$$

$\therefore (x-2)^2$  is a factor

so  $(x+3)^2$  is not a factor

**[B]**

Note: Multiple choice logic

If  $x+3$  was not a factor then  $(x+3)^2$  wouldn't be.

Similarly for  $x-2$  +  $(x-2)^2$ .

This means  $(x+3)$  +  $(x-2)$  are factors ... we can only eliminate one of the four options so it must be  $(x-2)^2$  or  $(x+3)^2$ .

$$2^2 \times (-3) = -12$$

$$2 \times (-3)^2 = 18$$

$\therefore (x+3)^2$  is NOT

$\therefore$  **[B]**

①

$$2/ \int \frac{\ln 2}{\sqrt{\pi - x^2}} dx$$

$$= \ln 2 \int \frac{1}{\sqrt{(\pi)^2 - (x)^2}} dx$$

Using  $\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1}\left(\frac{f(x)}{a}\right) + C$  we get

$$I = \ln 2 \cdot \sin^{-1} \frac{x}{\sqrt{\pi}} + C$$

$\therefore$  B

or differentiate each option

$$\text{eg A) } \ln 2 \times \left[ x \cdot \frac{\frac{1}{\pi}}{\sqrt{1 - \left(\frac{x}{\pi}\right)^2}} + \sin^{-1} \frac{x}{\pi} \cdot 1 \right]$$

$$= \ln 2 \cdot x \cdot \frac{\frac{1}{\pi}}{\frac{\sqrt{\pi^2 - x^2}}{\pi}} + \sin^{-1} \frac{x}{\pi}$$

$$= \frac{\ln 2}{\sqrt{\pi^2 - x^2}} + \sin^{-1} \frac{x}{\pi} \quad \times$$

etc

3. Normally for  $y = \cos^{-1} x$   $-1 \leq x \leq 1$ ,  $0 \leq y \leq \pi$

The range hasn't changed but now  $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

so

$$-1 \leq ? \leq 1$$

$$-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$$

$$-\pi \leq 4x \leq \pi$$

$$-1 \leq \frac{4}{\pi} x \leq 1$$

$$\therefore y = \cos^{-1}\left(\frac{4x}{\pi}\right)$$



4. You could choose

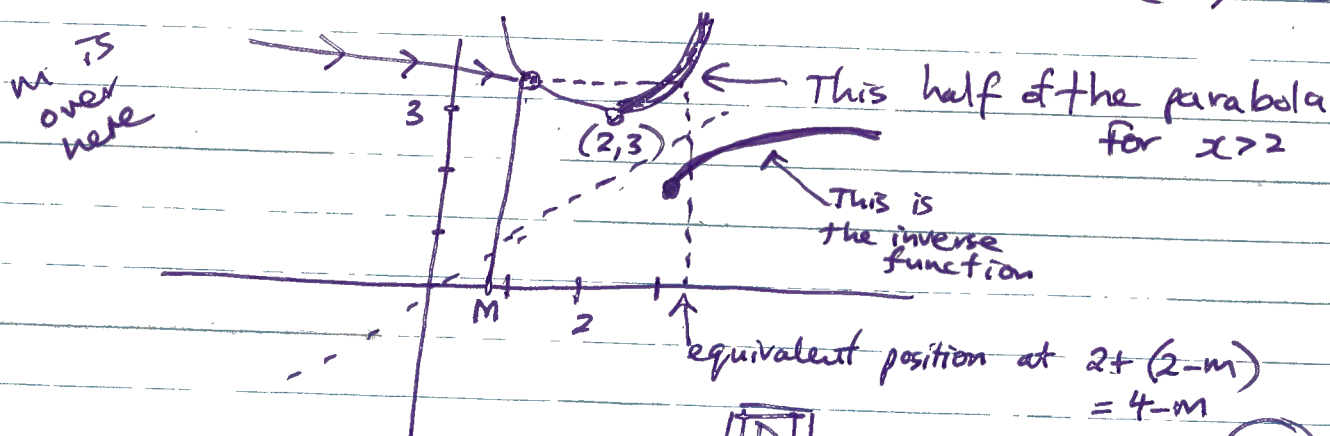
2 Red + 5 blue + 5 white + 5 green + 5 yellow  
and then the next marble will make  
6 of one colour.

$\therefore 23$



5.

$$x^2 - 4x + 7 = x^2 - 4x + 2^2 - 2^2 + 7 = (x-2)^2 + 3$$



3

$$6/ \quad (4)(3) + (-1)(m) = 0$$

$$m = 12$$

(dot product is zero for  $\perp$  vectors)



$$7/ \quad \frac{dT}{dt} = -k(T-15)$$

from 1<sup>st</sup> sentence + second dot point

$$\frac{dT}{T-15} = -k dt$$

$$\ln(T-15) = -kt + c$$

$$\log_e(T-15) = -kt + c$$

(note:  $\ln|T-15| = \ln(T-15)$  since  $T > 15$ )

$$e^{-kt+c} = T-15$$

$$me^{-kt} = T-15$$

(let  $m = e^c$ )

$$T = me^{-kt} + 15$$

when  $t = 5$   $T = 75$

when  $t = 8$   $T = 45$

$$75 = me^{-5k} + 15$$

$$45 = me^{-8k} + 15$$

$$60 = me^{-5k} \quad \text{--- (1)}$$

$$30 = me^{-8k} \quad \text{--- (2)}$$

$$\textcircled{1} \div \textcircled{2}$$

$$2 = \frac{e^{-5k}}{e^{-8k}}$$

$$2 = e^{3k}$$

$$\ln 2 = \ln e^{3k}$$

$$3k = \ln 2$$

$$k = \frac{\ln 2}{3}$$



8/  $\frac{dv}{dt}$  is the gradient

From the direction field the gradient is always negative eg when  $V=10$

Eliminate  $\rightarrow C + D$  are positive

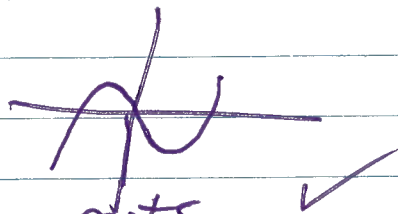
Now as  $V$  increases  $\frac{dv}{dt}$  is more negative (greater in size)

The opposite would happen for B

$\therefore$  A

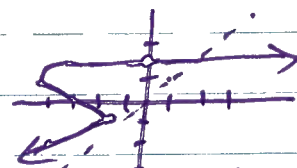
9/ A) Reflect in  $x$ -axis + make all points twice as far from  $x$ -axis  
 $\rightarrow$  still only 1 root

B) Translate curve up 3 units then



Yes 2 more roots ✓

C) Reflect in line  $y=x$



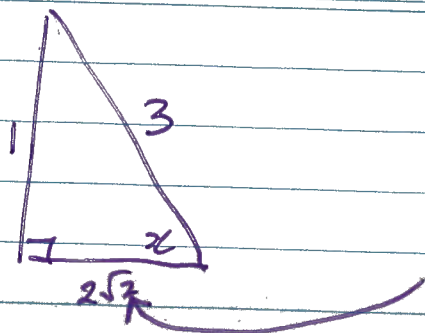
coords "swap"  $\therefore$  only root is at  $(-3, 0)$

D) Translate 3 units left .. still 1 root

$\therefore$  B

(5)

10/



By Pythagoras

$$\sqrt{3^2 - 1^2} = \sqrt{8} = 2\sqrt{2}$$

Now  $x$  is obtuse, so  $\cos x = -\frac{2\sqrt{2}}{3}$

$$\begin{aligned}\cos 2x &= \cos^2 x - \sin^2 x \\ &= \left(-\frac{2\sqrt{2}}{3}\right)^2 - \left(\frac{1}{3}\right)^2\end{aligned}$$

$$= \frac{8 - 1}{9}$$

$$= \frac{7}{9}$$

B

$$\text{1/ a) i) } \cos 7x \sin 5x = \frac{1}{2} [\sin(7x+5x) - \sin(7x-5x)] \\ = \frac{1}{2} [\sin 12x - \sin 2x]$$

$$\text{ii) } 2 \cos 7x \sin 5x = \sin 12x - \frac{\sqrt{3}}{2}$$

$$2 \times \frac{1}{2} [\sin 12x - \sin 2x] = \sin 12x - \frac{\sqrt{3}}{2}$$

$$\sin 12x - \sin 2x = \sin 12x - \frac{\sqrt{3}}{2}$$

$$\sin 2x = \frac{\sqrt{3}}{2}$$

$$2x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3}$$

$$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3}$$

b) Show true for  $n=1$

$$2^{2(1)} + 6(1) - 1 = 4 + 6 - 1 \\ = 9 \\ = 3 \times 3$$

$\therefore$  true for  $n=1$

Assume true for  $n=k$

$$\overset{ie}{2^{2k}} + 6k - 1 = 3M, \text{ M is an integer}$$

$$\text{rearranging } 2^k = 3M - 6k + 1 \quad *$$

Prove true for  $n=k+1$  if true for  $n=k$

ie RTP  $2^{k+1} + 6(k+1) - 1 = 3Q$   $Q$  is an integer

$$\begin{aligned} \text{LHS} &= 2^{2k+2} + 6k+6 - 1 \\ &= 2^{2k} \times 2^2 + 6k + 5 \\ &= (3M - 6k + 1) \times 4 + 6k + 5 \text{ from } * \end{aligned}$$

$$\begin{aligned} &= 12M - 24k + 4 + 6k + 5 \\ &= 12M - 18k + 9 \\ &= 3(4M - 6k + 3) \\ &= 3Q \text{ where } Q = 4M - 6k + 3 \\ &\text{and } Q \text{ is an integer since } 4, M, 6, k, 3 \text{ are integers} \end{aligned}$$

$\therefore 2^{2(k+1)} + 6(k+1) - 1$  is divisible by 3  
if  $2^k + 6k - 1$  is divisible by 3

Conclusion

Since  $2^{2(1)} + 6(1) - 1$  is divisible by 3, and  $2^{2(k+1)} + 6(k+1) - 1$  is divisible by 3 if  $2^k + 6k - 1$  is divisible by 3, by the principle of mathematical induction the statement is true for all integers  $\geq 1$ .

c)  $P(x) = x^3 + 0x^2 - 2px + q$

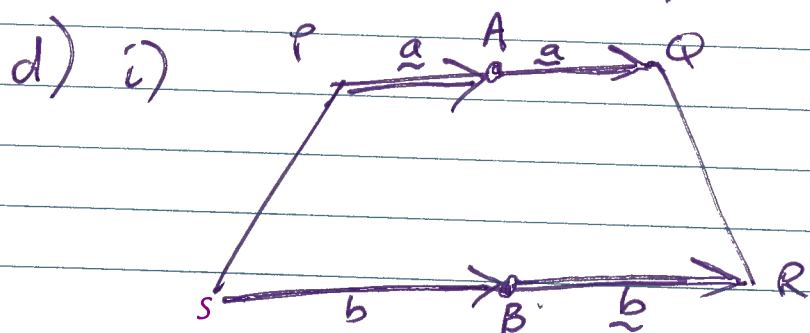
i) sum of roots is 0

$$\begin{aligned} \therefore \alpha + \beta + (\alpha + \beta) &= 0 \\ 2\alpha + 2\beta &= 0 \\ 2\alpha &= -2\beta \\ \alpha &= -\beta \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad \alpha\beta + \alpha(\alpha+\beta) + \beta(\alpha+\beta) &= -2p \\
 \alpha(-\alpha) + \alpha^2 + \alpha(-\alpha) + (-\alpha)(\alpha) + (-\alpha)(-\alpha) &= -2p \\
 -\alpha^2 + \alpha^2 - \alpha^2 - \alpha^2 + \alpha^2 &= -2p \\
 -\alpha^2 &= -2p \\
 \alpha^2 &= 2p \\
 \alpha &= \pm\sqrt{2p}
 \end{aligned}$$

$$\therefore \alpha + \beta = \sqrt{2p} - \sqrt{2p} = 0$$

The roots are  $-\sqrt{2p}$ ,  $\sqrt{2p}$  and 0.



$$\begin{aligned}
 \vec{QR} &= \vec{QA} + \vec{AB} + \vec{BR} \\
 &= -\underline{a} + \vec{AB} + \underline{b} \\
 &= \vec{AB} + \underline{b} - \underline{a}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad \vec{PS} &= \vec{PA} + \vec{AB} + \vec{BS} \\
 &= \underline{a} + \vec{AB} - \underline{b} \\
 &= \vec{AB} + (\underline{a} - \underline{b})
 \end{aligned}$$

$$\text{so } \vec{AB} = \vec{PS} - (\underline{a} - \underline{b}) \quad \dots \textcircled{1}$$

$$\text{and from i)} \quad \vec{AB} = \vec{QR} - (\underline{b} - \underline{a}) \quad \dots \textcircled{2}$$

$$\textcircled{1} + \textcircled{2}$$

$$2\vec{AB} = \vec{PS} + \vec{QR} - (\underline{a} - \underline{b}) - (\underline{b} - \underline{a}) \quad \textcircled{9}$$

$$2\vec{AB} = \vec{PS} + \vec{QR} \quad -a + b - b + a$$

$$2\vec{AB} = \vec{PS} + \vec{QR}$$

$$\vec{AB} = \frac{1}{2}(\vec{PS} + \vec{QR})$$

12/ i)  $R \sin(x+\alpha) = \sqrt{3} \sin x + 1 \cos x$

$$R \sin x \cos \alpha + R \cos x \sin \alpha = \sqrt{3} \sin x + 1 \cos x$$

$$(R \cos \alpha) \sin x + (R \sin \alpha) \cos x = \sqrt{3} \sin x + 1 \cos x$$

$$\therefore R \sin \alpha = 1 \quad \text{--- (1)}$$

$$R \cos \alpha = \sqrt{3} \quad \text{--- (2)}$$

$$\textcircled{1} \div \textcircled{2} \quad \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6} \quad \text{since } 0 < \alpha < \frac{\pi}{2}$$

$$\textcircled{1}^2 + \textcircled{2}^2 \quad R^2(\sin^2 \alpha + \cos^2 \alpha) = 1 + 3$$

$$R^2 = 4$$

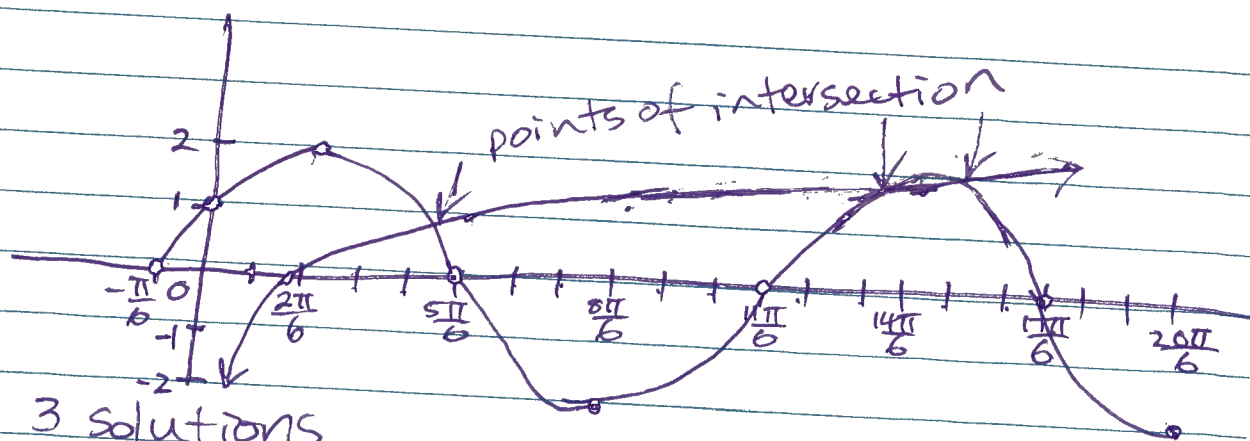
$$R = \pm 2 \quad \therefore R = 2 \quad \text{since } R > 0$$

so  $\sqrt{3} \sin x + \cos x = 2 \sin(x + \frac{\pi}{6})$

ii) Missing numbers are

1                      1.99                      2.19

iii)



iv) 3 solutions

(12)

$$b) (1+x)^m \Rightarrow {}^m C_r x^r$$

$$(1+x)^n \Rightarrow {}^n C_r x^r$$

so to get  $x^4$  we need on LHS

$${}^m C_0 x^0 {}^n C_4 x^4 + {}^m C_1 x^1 {}^n C_3 x^3 + {}^m C_2 x^2 {}^n C_2 x^2$$

$$+ {}^m C_3 x^3 {}^n C_1 x^1 + {}^m C_4 x^4 {}^n C_0 x^0$$

ie  $({}^m C_0 {}^n C_4 + {}^m C_1 {}^n C_3 + {}^m C_2 {}^n C_2 + {}^m C_3 {}^n C_1 + {}^m C_4 {}^n C_0) x^4$

On RHS

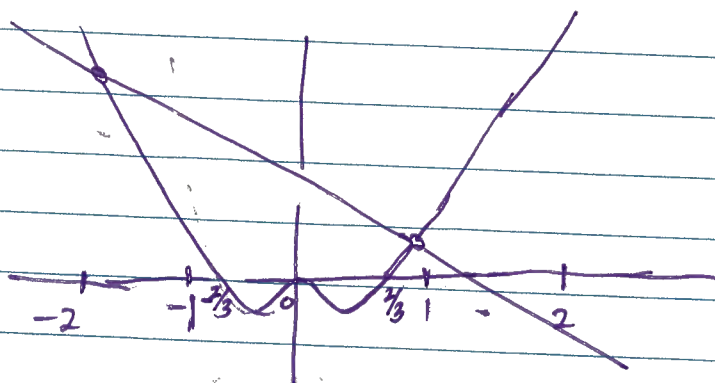
$${}^{m+n} C_4 x^4$$

$$\therefore {}^{m+n} C_4 = {}^m C_0 {}^n C_4 + {}^m C_1 {}^n C_3 + {}^m C_2 {}^n C_2 + {}^m C_3 {}^n C_1 + {}^m C_4 {}^n C_0$$

$$= {}^n C_4 + {}^m C_1 {}^n C_3 + {}^m C_2 {}^n C_2 + {}^m C_3 {}^n C_1 + {}^m C_4$$

$$= {}^n C_4 + {}^n C_3 {}^m C_1 + {}^n C_2 {}^m C_2 + {}^n C_1 {}^m C_3 + {}^m C_4$$

c) i)



for x-intercepts

$$3x^3 - 2x^2 = 0$$

$$x^2(3x-2) = 0$$

$$x = 0, \frac{2}{3}$$

$\therefore$  by reflection  $-\frac{2}{3}$

ii)

$$y = 3(1)^3 - 2(1)^2 = 1 \Rightarrow \text{intersect at } (1, 1)$$

$$y = 3(-2)^3 - 2(-2)^2 = 16 \Rightarrow \text{intersect at } (-2, 16)$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{16 - 1}{-2 - 1} (x - 1)$$

$$y - 1 = -5(x - 1)$$

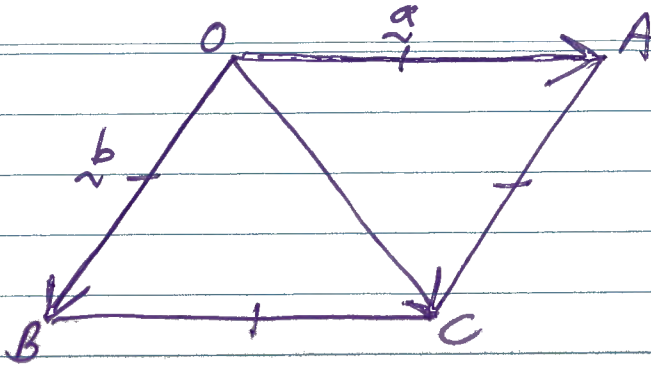
$$y = -5x + 5 + 1$$

$$y = -5x + 6$$

$$\therefore m = -5, b = 6$$

$$d) \quad 6 \times 5 \times 4 = 120 \quad ({}^6P_3)$$

13/



$$\begin{aligned}\vec{OC} &= \vec{OA} + \vec{AC} \\ &= \vec{OA} + \vec{OB} \\ &= \underline{a} + \underline{b}\end{aligned}$$

$$\begin{aligned}\vec{OC} \cdot \vec{OA} &= (\underline{a} + \underline{b}) \cdot \underline{a} \\ &= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} \\ &= |\underline{a}|^2 + |\underline{a}||\underline{b}|\cos\angle AOB\end{aligned}$$

but  $|\underline{a}| = |\underline{b}| \therefore \vec{OC} \cdot \vec{OA} = |\underline{a}|^2 + |\underline{a}|^2 \cos\angle AOB$

$$\begin{aligned}\vec{OC} \cdot \vec{OB} &= (\underline{a} \cdot \underline{b}) \cdot \underline{b} \quad \text{①} \\ &= \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{b} \\ &= |\underline{a}||\underline{b}|\cos\angle AOB + |\underline{b}|^2\end{aligned}$$

but  $|\underline{a}| = |\underline{b}| \therefore \vec{OC} \cdot \vec{OB} = |\underline{a}|^2 + |\underline{a}|^2 \cos\angle AOB$  ②

From ① + ②

$$\begin{aligned}\vec{OC} \cdot \vec{OA} &= \vec{OC} \cdot \vec{OB} \\ |\vec{OC}||\underline{a}|\cos\angle AOC &= |\vec{OC}||\underline{b}|\cos\angle BOC\end{aligned}$$

but  $|\underline{a}| = |\underline{b}|$

13

$$|\vec{OC}| |\vec{a}| \cos LAOC = |\vec{OC}| |\vec{a}| \cos LBOC$$

$$\therefore \cos LAOC = \cos LBOC$$

Now both LAOC and LBOC are in the range from 0 to  $180^\circ$ .

$$\therefore LAOC = LBOC$$

$$\begin{aligned} \text{b)} \quad V &= \pi \int_0^h \left( \frac{3}{\sqrt{9+x^2}} \right)^2 dx \\ &= \pi \int_0^h \frac{9}{9+x^2} dx \\ &= 9\pi \int_0^h \frac{1}{9+x^2} dx \\ &= 9\pi \left[ \frac{1}{3} \tan^{-1} \frac{x}{3} \right]_0^h \\ &= 3\pi \left[ \tan^{-1} \frac{h}{3} - \tan^{-1} \frac{0}{3} \right] \\ &= 3\pi \tan^{-1} \left( \frac{h}{3} \right) \quad \text{cubic units} \end{aligned}$$

$$\text{c)} \quad \frac{dh}{dt} = 3 \text{ cm/s} \quad \text{find } \frac{dv}{dt} \text{ when } h=6$$

$$\begin{aligned} V &= 3\pi \tan^{-1} \left( \frac{h}{3} \right) \\ \frac{dv}{dh} &= 3\pi \cdot \frac{\frac{1}{3}}{1 + \left( \frac{h}{3} \right)^2} \end{aligned}$$

$$\frac{dv}{dh} = 3\pi \frac{\frac{3}{2}}{\frac{9+h^2}{9}}$$

$$= \frac{9\pi}{9+h^2}$$

$$\text{When } h=6 \quad \frac{dv}{dh} = \frac{9\pi}{9+6^2} = \frac{9\pi}{45} = \frac{\pi}{5}$$

$$\begin{aligned} \text{Now } \frac{dv}{dt} &= \frac{dv}{dh} \times \frac{dh}{dt} \\ &= \frac{\pi}{5} \times 3 \\ &= 0.6\pi \text{ cm}^3/\text{s} \end{aligned}$$

$$\begin{aligned} \text{c) } S &= \frac{2000}{1+199e^{-0.4t}} \\ &= 2000(1+199e^{-0.4t})^{-1} \end{aligned}$$

$$\frac{ds}{dt} = 2000 \times (-1) \times 199 \times (-0.4)e^{-0.4t} \times (1+199e^{-0.4t})^{-2}$$

$$= \frac{(-2000 \times 199 \times -0.4)e^{-0.4t}}{(1+199e^{-0.4t})^2}$$

$$= \frac{2000 \times (0.4 \times 199 e^{-0.4t})}{(1+199e^{-0.4t}) \times (1+199e^{-0.4t})}$$

$$= S \times \frac{\frac{2}{5} \times 199 e^{-0.4t}}{(1+199e^{-0.4t})}$$

$$= \frac{S}{5} \left[ \frac{2 \times 199 e^{-0.4t}}{1+199e^{-0.4t}} \right]$$

$$= \frac{S}{5} \left[ \frac{2 \times 1000}{1000} \times \frac{199e^{-0.4t}}{1+199e^{-0.4t}} \right]$$

$$= \frac{S}{5} \left[ \frac{2000}{1+199e^{-0.4t}} \times \frac{199e^{-0.4t}}{1000} \right]$$

$$= \frac{S}{5} \left[ S \times \frac{199e^{-0.4t}}{1000} \right]$$

$$= \frac{S}{5} \left[ \frac{S}{1000} \times 199e^{-0.4t} \right]$$

Now, from  $S = \frac{2000}{1+199e^{-0.4t}}$  we get

$$S(1+199e^{-0.4t}) = 2000$$

$$1+199e^{-0.4t} = \frac{2000}{S}$$

$$199e^{-0.4t} = \frac{2000}{S} - 1$$

$$= \frac{S}{5} \left[ \frac{S}{1000} \times \left( \frac{2000}{S} - 1 \right) \right]$$

$$= \frac{S}{5} \left[ 2 - \frac{S}{1000} \right]$$

ii) When  $t=0$ ,  $S = \frac{2000}{1+199} = 10$

as  $t \rightarrow -\infty$   $S \rightarrow 0$

as  $t \rightarrow \infty$   $S \rightarrow 2000 \therefore B$

( $t=0, 2000$  are given by horizontal slope line segments since they are constant solutions.)

iii) When does  $\frac{ds}{dt}$  reach a maximum?

$$\begin{aligned}\frac{ds}{dt} &= \frac{s}{5} \left( 2 - \frac{s}{1000} \right) \\ &= \frac{2s}{5} - \frac{s^2}{5000}\end{aligned}$$

$$= -\frac{s^2}{5000} + \frac{2s}{5}$$

$$= \left( -\frac{1}{5000} \right) s^2 + \frac{2}{5} s$$

This is a parabola (quadratic)  
Since  $a < 0$  it is concave down.

$$\text{Max when } S = -\frac{b}{2a} = \frac{-\frac{2}{5}}{2\left(-\frac{1}{5000}\right)}$$

$$= -\frac{2}{5} \div -\frac{2}{5000}$$

$$= -\frac{2}{5} \times \frac{5000}{-2}$$

$$= 1000$$

$$\text{When } S = 1000, \quad 1000 = \frac{2000}{1 + 199e^{-0.4t}}$$

$$1 + 199e^{-0.4t} = 2$$

$$199e^{-0.4t} = 1$$

$$e^{-0.4t} = \frac{1}{199}$$

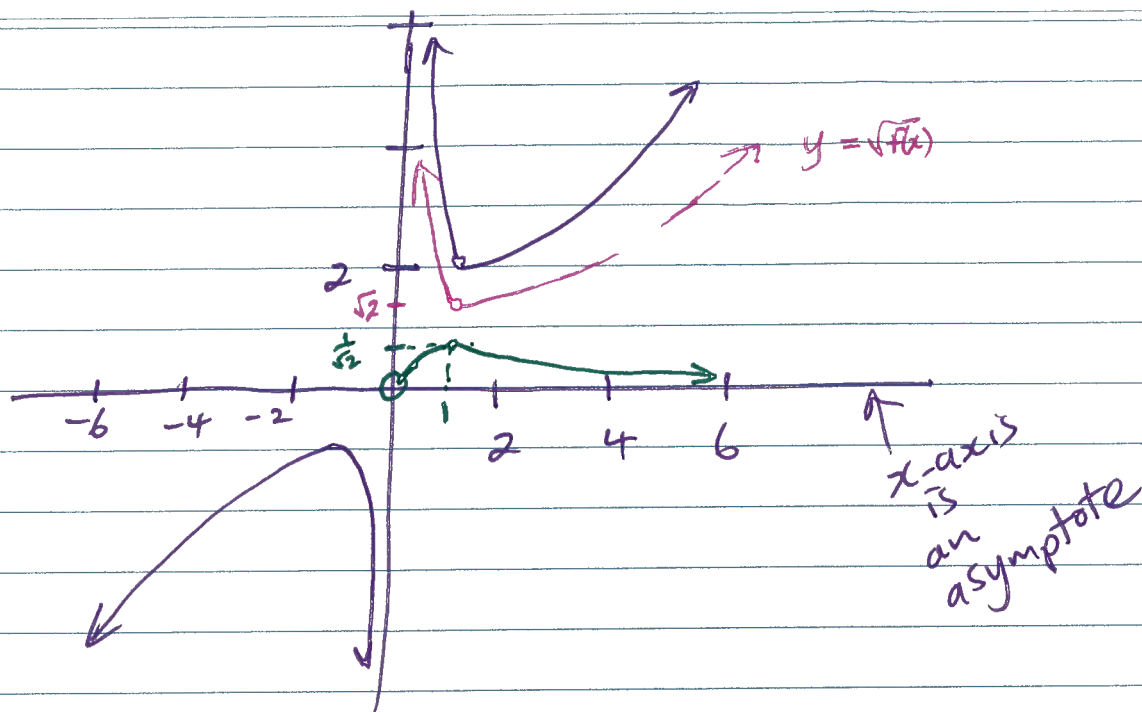
$$e^{0.4t} = 199$$

$$0.4t = \ln 199$$

$$t = \frac{\ln 199}{0.4} = 13.233 \quad (17)$$

About  
13 days

14/



$$f(x) = \frac{x^2+1}{x}$$

$$f'(x) = \frac{x \cdot 2x - (x^2+1) \cdot 1}{x^2} = \frac{2x^2 - x^2 - 1}{x^2} = \frac{x^2 - 1}{x^2}$$

$$\text{for T.P. } \frac{x^2 - 1}{x^2} = 0$$

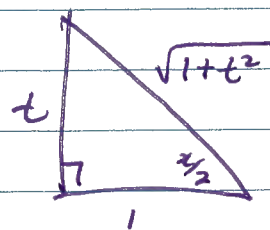
$$\therefore x = 1 \text{ or } -1$$

Can't  $\sqrt{\text{negative}}$  so we won't consider LHS of y-axis part of original curve.

$\therefore x=1$  gives min T.P. on original & max T.P. on final curve.

b) i) let  $t = \tan \frac{x}{2}$

$$\begin{aligned} \therefore \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} \\ &= \frac{1}{2} \left( \frac{\sqrt{1+t^2}}{1} \right)^2 \\ &= \frac{1+t^2}{2} \end{aligned}$$



$$\begin{aligned} \frac{dx}{dt} &= \frac{2}{1+t^2} \\ dx &= \frac{2dt}{1+t^2} \end{aligned}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$\begin{aligned} \int \frac{1}{5+3\cos x} dx &= \int \frac{1}{5+3\frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2} \\ &= \int \frac{1}{\frac{5+5t^2+3-3t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2} \\ &= \int \frac{1}{8+2t^2} \cdot 2dt \\ &= \int \frac{1}{4+t^2} \cdot dt \\ &= \frac{1}{2} \tan^{-1} \frac{t}{2} + C \\ &= \frac{1}{2} \tan^{-1} \left( \frac{\tan \frac{x}{2}}{2} \right) + C \\ &= \frac{1}{2} \tan^{-1} \left( \frac{1}{2} \tan \left( \frac{x}{2} \right) \right) + C \end{aligned}$$

$$\begin{aligned}
 \text{ii) Area} &= \int_0^{\pi} \frac{1}{5+3\cos x} \cdot dx \\
 &= \left[ \frac{1}{2} \tan^{-1} \left( \frac{1}{2} \tan \frac{x}{2} \right) \right]_0^{\pi} \\
 &= \frac{1}{2} \left[ \tan^{-1} \left( \frac{1}{2} \tan \frac{\pi}{2} \right) - \tan^{-1} \left( \frac{1}{2} \tan \frac{0}{2} \right) \right] \\
 &= \frac{1}{2} \left[ \frac{\pi}{2} - 0 \right] \\
 &= \frac{\pi}{4}
 \end{aligned}$$

But  $\tan \frac{\pi}{2}$  is actually undefined.  
 So:

$$\begin{aligned}
 A &= \int_0^{\infty} \frac{1}{4+t^2} dt \quad \leftarrow \text{when } x=\pi, t=\tan \frac{x}{2} \\
 &= \left[ \frac{1}{2} \tan^{-1} \left( \frac{t}{2} \right) \right]_0^{\infty} \\
 &= \frac{1}{2} \tan^{-1}(\infty) - \frac{1}{2} \tan^{-1} 0 \\
 &= \frac{1}{2} \left( \frac{\pi}{2} \right) - 0 \\
 &= \frac{\pi}{4}
 \end{aligned}$$

$$c) i) \quad \underline{a}(t) = 0 \underline{i} - 10 \underline{j}$$

$$\underline{v}(t) = \int \underline{a}(t) \cdot dt$$

$$\underline{v}(t) = -10t \underline{j} + c$$

$$\text{when } t=0, \quad \underline{v}(t) = 90 \cos 30^\circ \underline{i} + 90 \sin 30^\circ \underline{j}$$

$$= 90 \cdot \frac{\sqrt{3}}{2} \underline{i} + 90 \cdot \frac{1}{2} \underline{j}$$

$$= 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$\text{so } \underline{v}(0) = -10t \underline{j} + c = 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$-10(0) \underline{j} + c = 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$c = 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$\underline{v}(t) = -10t \underline{j} + 45\sqrt{3} \underline{i} + 45 \underline{j}$$

$$\underline{v}(t) = 45\sqrt{3} \underline{i} + (45-10t) \underline{j}$$

$$\underline{s}(t) = \int \underline{v}(t) dt$$

$$= \int [45\sqrt{3} \underline{i} + (45-10t) \underline{j}] dt$$

$$= 45\sqrt{3}t \underline{i} + (45t - 5t^2) \underline{j} + c$$

$$\text{when } t=0, \quad \underline{s}(t) = 0 \underline{i} + 0 \underline{j}$$

$$\text{so } \underline{s}(0) = 45\sqrt{3}(0) \underline{i} + (45(0) - 5(0)^2) \underline{j} + c = 0 \underline{i} + 0 \underline{j}$$

$$c = 0$$

$$\therefore \underline{s}(t) = 45\sqrt{3}t \underline{i} + (45t - 5t^2) \underline{j} \quad (21)$$

ii) For the stone  $\underline{a}(t) = -10 \underline{j}$

$$\underline{x}(t) = V \underline{i} - 10t \underline{j}$$

$$\underline{s}(t) = Vt \underline{i} - 5t^2 \underline{j} + 20 \underline{j}$$

$$\underline{s}(t) = Vt \underline{i} + (20 - 5t^2) \underline{j}$$

Objects collide when position vectors are equal

BUT the stone wasn't fired until 5 seconds after the golf ball was hit.  
 $\therefore$  for the stone, relative to the golf-ball

$$\underline{s}(t) = V(t-5) \underline{i} + (20 - 5(t-5)^2) \underline{j}$$

Now we can equate the position vectors of the golf-ball and the stone

$$V(t-5) \underline{i} + (20 - 5(t-5)^2) \underline{j} = (45\sqrt{3}t) \underline{i} + (45t - 5t^2) \underline{j}$$

$$\therefore V(t-5) = 45\sqrt{3}t \text{ and } 20 - 5(t-5)^2 = 45t - 5t^2$$

(equating components)

Using the  $\underline{j}$  components:

$$20 - 5(t-5)^2 = 45t - 5t^2$$

$$20 - 5[t^2 - 10t + 25] = 45t - 5t^2$$

$$20 - 5t^2 + 50t - 125 = 45t - 5t^2$$

$$50t - 105 = 45t$$

$$5t = 105$$

$$t = 21 \text{ s}$$

iii) When  $t=21$  the  $\hat{i}$  components of the position vectors are equal

$$\therefore V(t-5) = 45\sqrt{3} t$$

$$V(21-5) = 45\sqrt{3} (21)$$

$$V = \frac{(45\sqrt{3}(21))}{16}$$

$$V = \frac{945\sqrt{3}}{16} \text{ ms}^{-1}$$

So for the stone

$$V(t) = V\hat{i} - 10t\hat{j}$$

$$V(t) = \frac{945\sqrt{3}}{16} \hat{i} - 10t\hat{j}$$

$$V(16) = \frac{945\sqrt{3}}{16} \hat{i} - 10(16)\hat{j}$$

$$V(16) = \frac{945\sqrt{3}}{16} \hat{i} - 160\hat{j}$$

$$\text{The speed is } |V(21)| = \sqrt{\left(\frac{945\sqrt{3}}{16}\right)^2 + (160)^2}$$

$$= 189.9$$

190  $\text{ms}^{-1}$  to 3 significant figures.